

EE446 Lab 6
Zelamir Reynolds-Henkins

Part 1

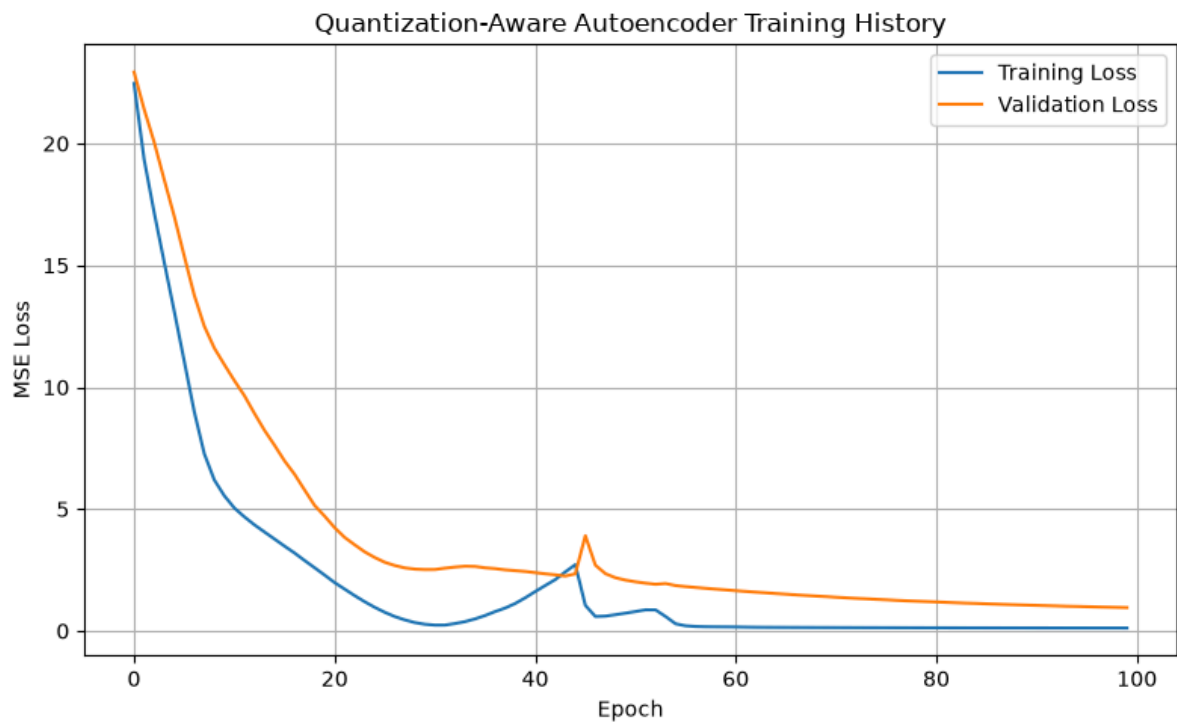
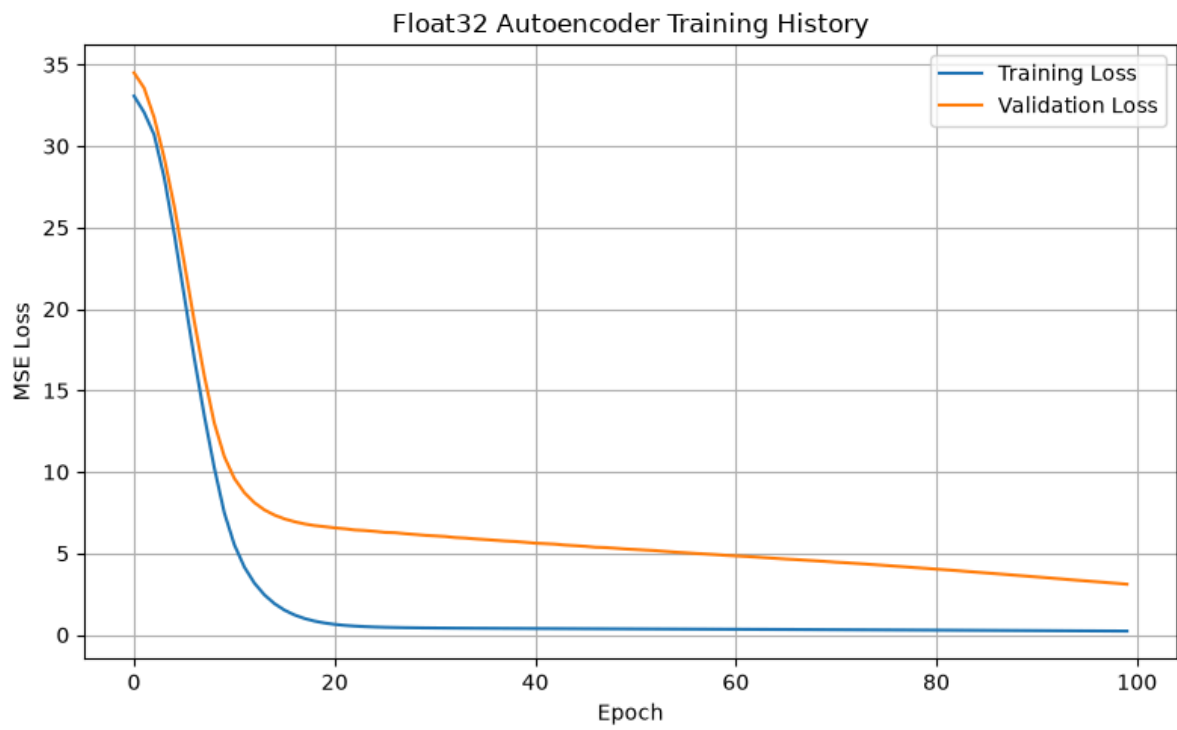
Edge Impulse Link: <https://studio.edgeimpulse.com/public/1073998/live>

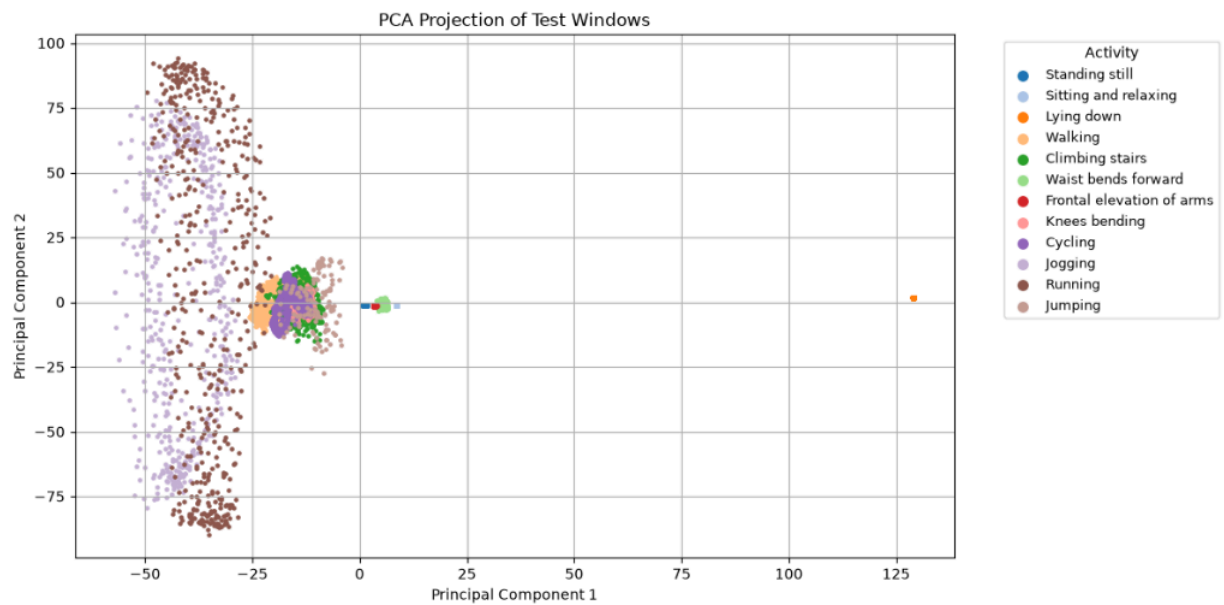
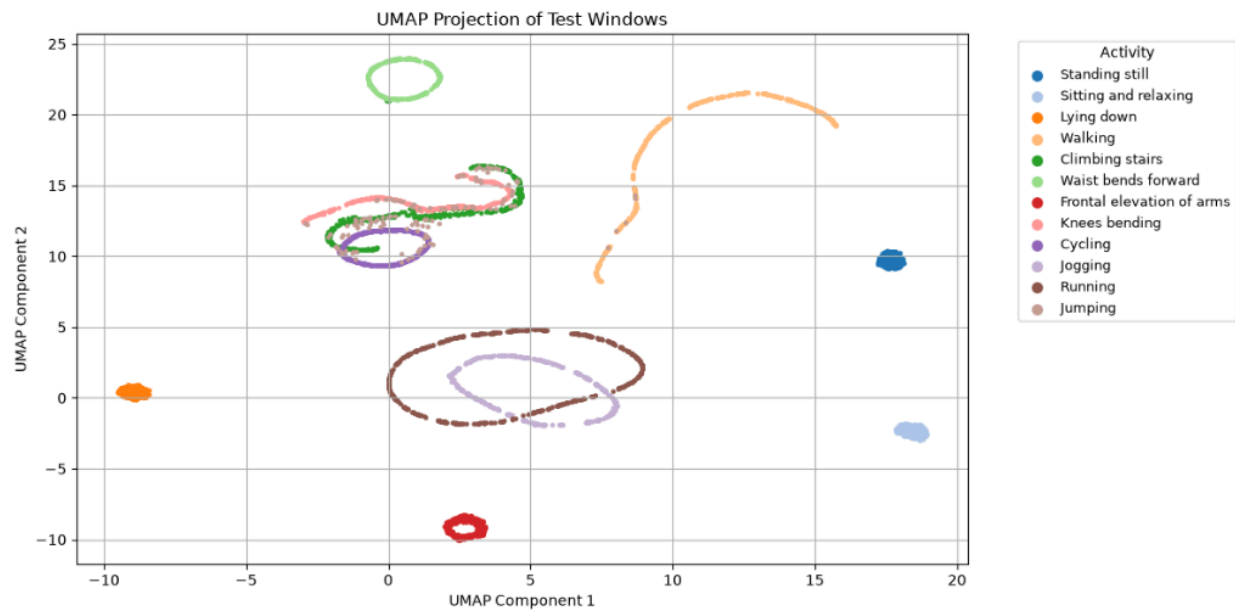
```
C:\WINDOWS\system32\cmd. x + v
Press any key to continue . . .

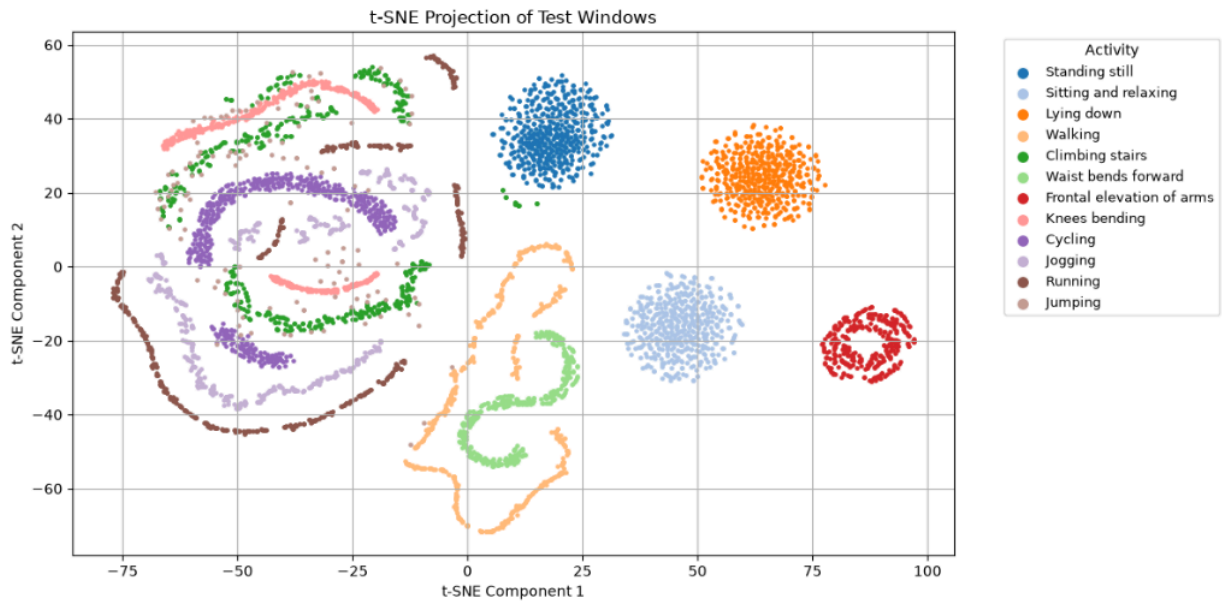
C:\Users\zelam\Downloads\arduino-cli_1.5.2-rc.1_Windows_64bit (1)>edge-impulse-run-impulse
Edge Impulse impulse runner v1.39.2
? Which device do you want to connect to? ( type to search) COM20
[SER] Connecting to COM20
[SER] Serial is connected, trying to read config...
Failed to parse snapshot line [ ]
[SER] Retrieved configuration
[SER] Device is running AT command version 1.8.0
[SER] Started inferencing, press CTRL+C to stop...
LSE
Inferencing settings:
    Interval: 10.00 ms.
    Frame size: 600
    Sample length: 2000 ms.
    No. of classes: 2
Starting inferencing, press 'b' to break
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 609 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
    nominal: 0.039063
    off: 0.960937
Anomaly prediction: 9.963060
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 610 us, anomaly 506 us, postprocessing 49 us
#Classification predictions:
    nominal: 0.437500
    off: 0.562500
Anomaly prediction: 15.883278
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 610 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
    nominal: 0.937500
    off: 0.062500
Anomaly prediction: 15.000834
Starting inferencing in 2 seconds...
Sampling...
```

```
C:\WINDOWS\system32\cmd. X + v
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 592 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.195312
  off: 0.804687
Anomaly prediction: 451.139618
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 610 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.261719
  off: 0.738281
Anomaly prediction: 239.451096
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 612 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.597656
  off: 0.402344
Anomaly prediction: 371.471771
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 610 us, anomaly 6 ms, postprocessing 66 us
#Classification predictions:
  nominal: 0.996094
  off: 0.000000
Anomaly prediction: 19.040779
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 611 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 15.900782
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 593 us, anomaly 524 us, postprocessing 50 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
```

Part 2





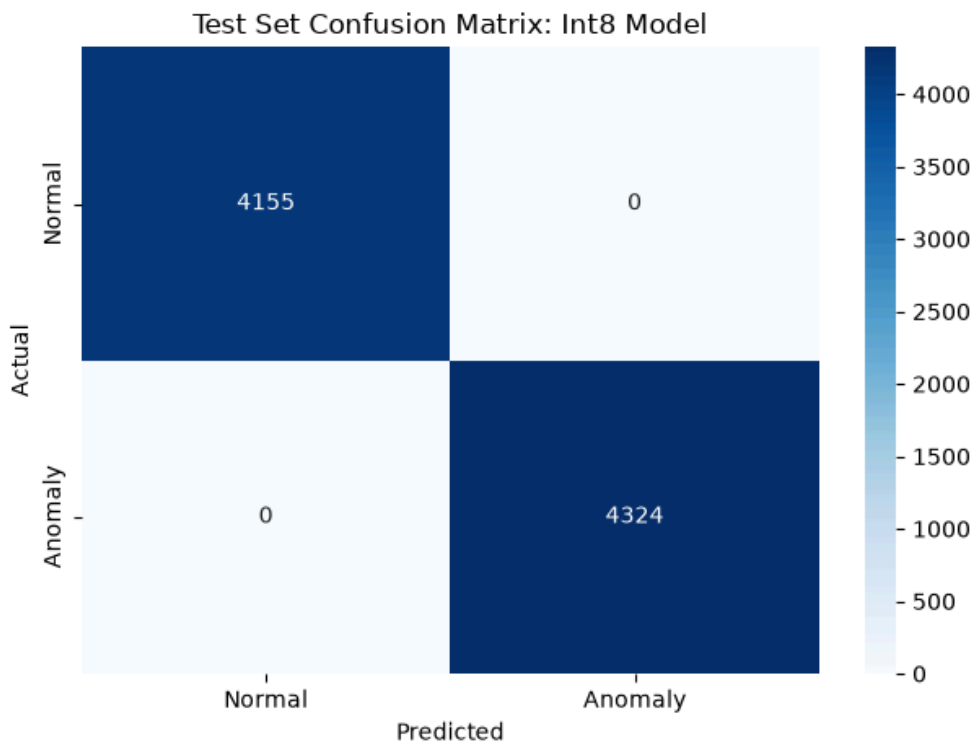
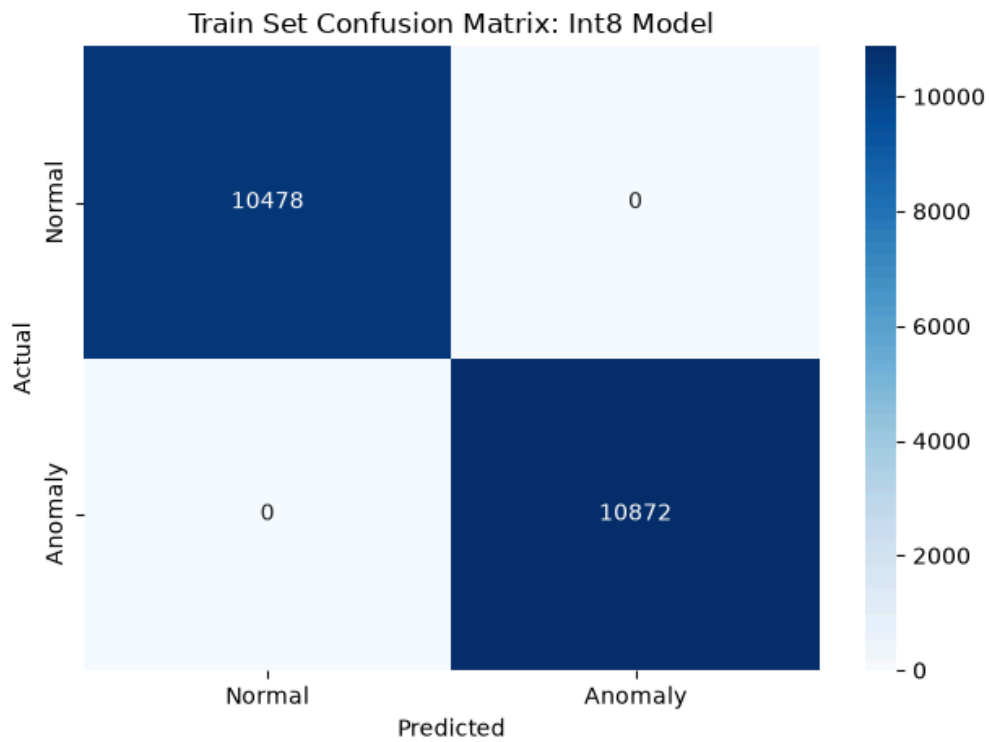


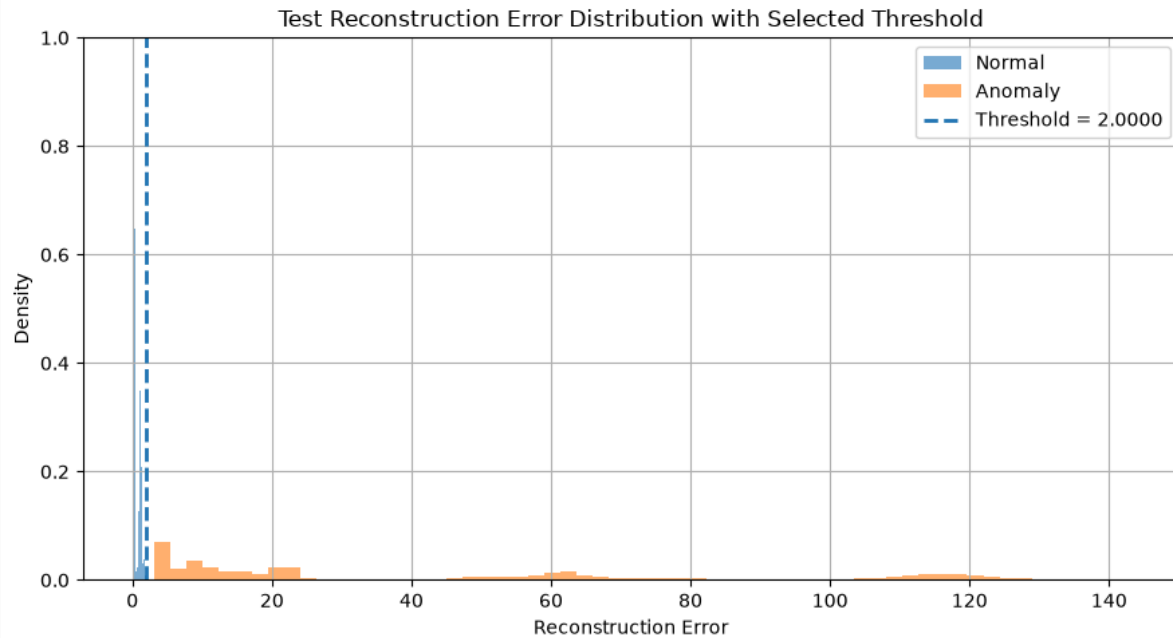
Train set evaluation: int8 model

	precision	recall	f1-score	support
Normal (0)	1.00	1.00	1.00	10478
Anomaly (1)	1.00	1.00	1.00	10872
accuracy			1.00	21350
macro avg	1.00	1.00	1.00	21350
weighted avg	1.00	1.00	1.00	21350

Test set evaluation: int8 model

	precision	recall	f1-score	support
Normal (0)	1.00	1.00	1.00	4155
Anomaly (1)	1.00	1.00	1.00	4324
accuracy			1.00	8479
macro avg	1.00	1.00	1.00	8479
weighted avg	1.00	1.00	1.00	8479





Threshold set to 2!

Output Serial Monitor X

Message (Enter to send message to 'Arduino Nano 33 BLE' on '...)

```
IMU initialized.
Allocating tensors.
AllocateTensors successful.
Model setup complete.
Reconstruction error: 0.200501
Result: Normal activity
Reconstruction error: 0.200493
Result: Normal activity
Reconstruction error: 0.200485
Result: Normal activity
Reconstruction error: 0.200491
Result: Normal activity
Reconstruction error: 0.200478
Result: Normal activity
Reconstruction error: 0.200495
Result: Normal activity
Reconstruction error: 0.200491
Result: Normal activity
Reconstruction error: 0.200501
```